

Efficient 3D Biomedical Image Segmentation by Parallely Multiscale Transformer–CNN Aggregation Network

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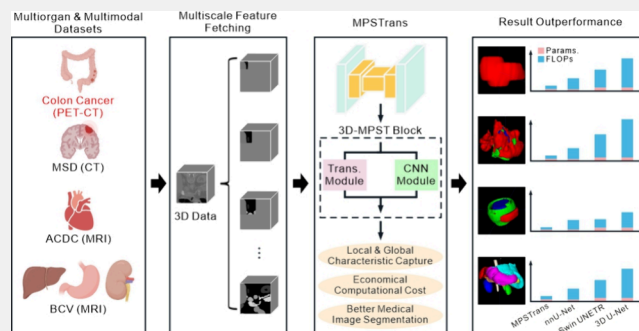
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ABSTRACT: Accurate and automated segmentation of 3D biomedical images is a sophisticated imperative in clinical diagnosis, imaging-guided surgery, and prognosis judgment. Although the burgeoning of deep learning technologies has fostered smart segmentors, the successive and simultaneous garnering global and local features still remains challenging, which is essential for an exact and efficient imageological assay. To this end, a segmentation solution dubbed the mixed parallel shunted transformer (MPSTrans) is developed here, highlighting 3D-MPST blocks in a U-form framework. It enabled not only comprehensive characteristic capture and multiscale slice synchronization but also deep supervision in the decoder to facilitate the fetching of hierarchical representations. Performing on an unpublished colon cancer data set, this model achieved an impressive increase in dice similarity coefficient (DSC) and a 1.718 mm decrease in Hausdorff distance at 95% (HD95), alongside a substantial shrink of computational load of 56.7% in giga floating-point operations per second (GFLOPs). Meanwhile, MPSTrans outperforms other mainstream methods (Swin UNETR, UNETR, nnU-Net, PHTrans, and 3D U-Net) on three public multiorgan (aorta, gallbladder, kidney, liver, pancreas, spleen, stomach, etc.) and multimodal (CT, PET-CT, and MRI) data sets of medical segmentation decathlon (MSD) brain tumor, multiatlas labeling beyond cranial vault (BCV), and automated cardiac diagnosis challenge (ACDC), accentuating its adaptability. These results reflect the potential of MPSTrans to advance the state-of-the-art in biomedical imaging analysis, which would offer a robust tool for enhanced diagnostic capacity.

KEYWORDS: 3D biomedical image segmentation, shunted transformer, convolutional neural networks, parallel architecture, multiscale feature extraction



INTRODUCTION

Medical image segmentation (MIS) aims to identify specific organs or lesions within human tissues, which is vital for disease diagnosis,¹ radiation therapy,² and preoperative planning.³ Conventional medical imageology relies on experienced practitioners to peruse those areas of interest frame by frame, which is labor-intensive and time-consuming especially in the three-dimensional (3D) cases. Fortunately, such a burden on healthcare professionals is lifted by recourse to automated MIS solutions, which not only streamline the standard operating procedures but suppress subjective errors, thus conducive to correct and coherent surgical decision-making.^{4,5} In the light of this significance, qualified algorithms are in dire need for scenario-specific 3D MIS. In the evolution of computational bioimaging techniques, the convolutional neural networks (CNNs) on behalf of the Deep Learning frameworks have become the mainstay due to their knack in

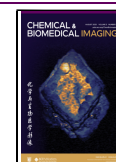
spatial signature extraction.^{6–8} Besides, U-Net, as another representative architecture in U-shape, could concatenate an encoder-decoder manifold to catch contextual information with supplemental segmentation grits,⁹ as well as better localization via skip connectors to cascade the down- to upsampling pathway. Despite the already amenable CNN-based builds for diverse MIS tasks, they still struggle to fathom long-range dependence in 3D medical images, which is vital for precision profiling tumors of size variance.¹⁰ To address this issue, upgraded 3D U-Nets were developed, though being

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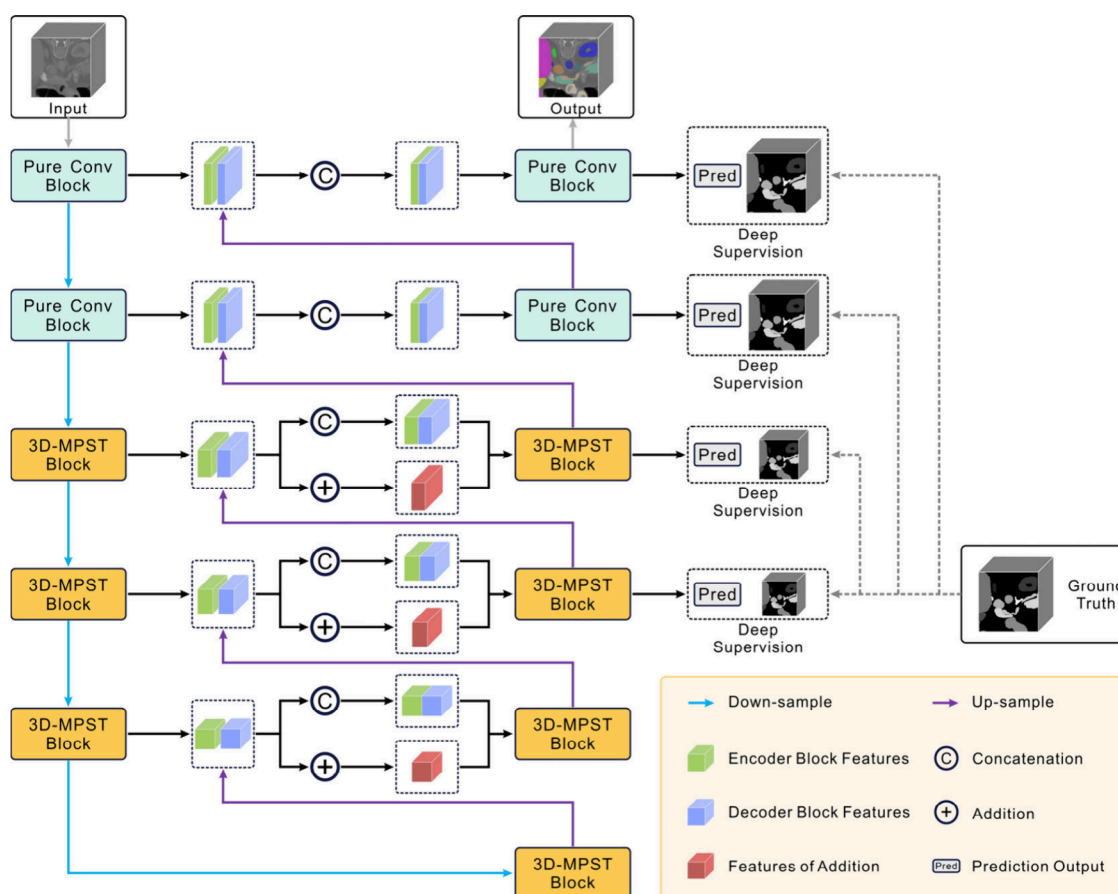


Figure 1. Hierarchy of MPSTrans. It signifies serial Pure Conv Blocks for initial feature fetching and 3D-MPST Blocks for spatial refinement. The ensemble employs downsampling in the encoder for characteristic compression and upsampling in the decoder for picture reconstruction. Features along the cascading encoder route are delivered to the counterpart decoder node for skip-connecting integration and superaddition of Trans-CNN features. The Deep Supervision wing outputs prediction by virtue of cross-entropy and dice loss functions for training optimization.

capable of evaluating volumetric data in an accurate way, yet less generalizable.^{11,12}

Recently, the research on the combination of CNNs and transformer (Trans) have attracted broad attention in the field of computer vision. To exemplify, frameworks like TransU-Net and SwinU-net employ Trans blocks as deep encoders or decoders to collect universal information in multisplanchic segmentation,^{11,12} while models like Swin UNTER,¹³ Cotr,¹⁴ TransU-Net, and SAM have successfully showcased the sensitive grasps of both local and global (L&G) traits.^{15–17} Nonetheless, challenges persist in striking the subtle balances between L&G feature extraction, which is essential for augmenting the accuracy of MIS. This is due to the fact that biomedical visualizations share characteristics of elusive edges plus blurry backgrounds, leading to difficulties in boundary discrimination. Although CNNs boast a proficiency in local hallmark mining, their receptive fields are restricted, potentially resulting in a deficiency of global semantics. As a solution, increasing the network depth could fortify the feature-extracting capacity but may also bring forth gradient vanishing or even explosion, needless to say the risk of overfitting, capturing more noise, or causing detail loss.¹⁸ Alternatively, Trans models on the basis of U-shaped structures, such as TransU-Net and Swin UNTER, utilize self-attention mechanisms to elicit global contextual info, meanwhile maintaining multiscale details through skip connections. However, balancing features at various levels with global ones still

poses a problem. Furthermore, the computational complexity and resource requirements of Trans-based schemes also make a rigid roadblock toward their clinical applications.

In view of the aforementioned, herein we propose a mixed parallel shunted Trans (dubbed MPSTrans) flow (Figure 1), which highlights the integration of Shunted Trans (STrans) with CNNs into a single block to simultaneously survey L&G landscapes for 3D MIS. The standard STrans is responsible for retrieving 3D patches from the volume to derive a 3D-distributed self-attention mechanism,¹⁹ of which learning progress the clarity and transparency can be improved by further incorporating deep supervision in the decoder. To be specific, the main contributions of this work are as follows: (1) building up a cohesive 3D-MPSTrans block that comprises a couple of 3D-STrans and CNN modules in parallel, which layout is conducive to concurrent withdrawal of multileveled meaningful attributes along their progressive pooling, leveraging complementary strengths of Trans with macroscopic and CNNs with microscopic processability, thus effectively gathering both global and local knowledge, (2) developing a 3D MIS strategy, which is able to scan and scrutinize cross-dimensional tumors and tissues out of 3D biomedical imaging for both panoramic intelligence and partial insights, (3) conducting extensive experiments on multimodal data sets, which involves computed tomography (CT), positron emission tomography (PET)-CT, and magnetic resonance imaging (MRI) in company with data sets regarding an

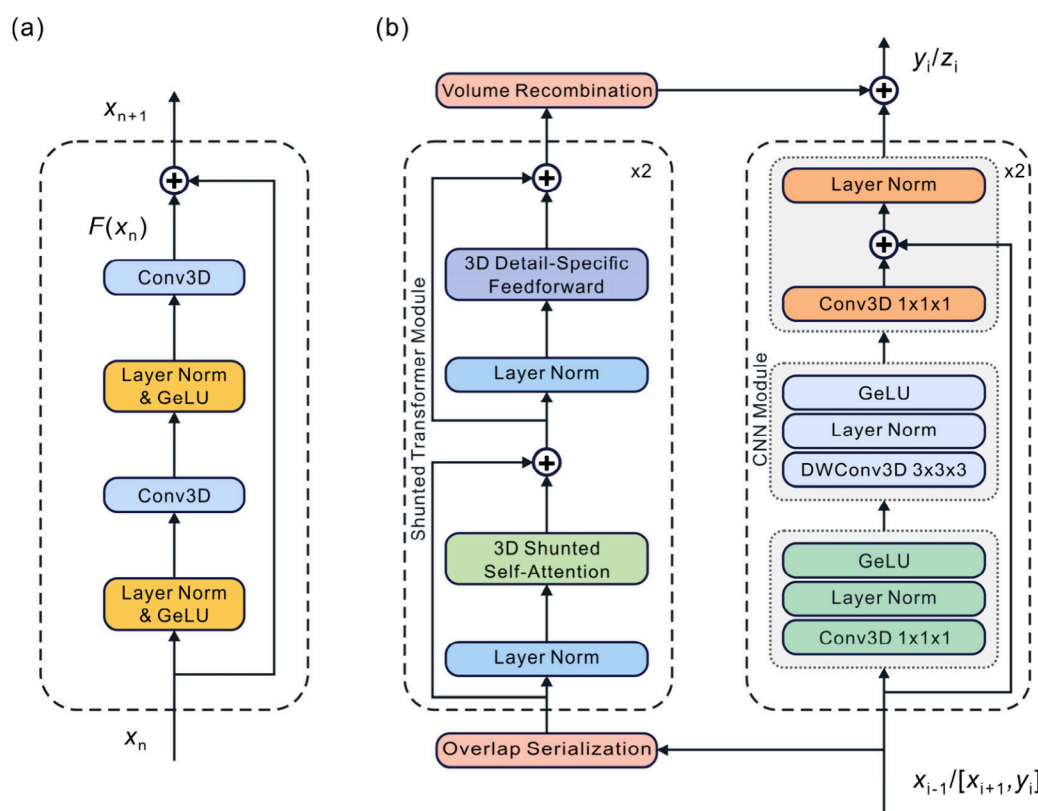


Figure 2. (a) Loop of Pure Conv Block, in which x_n and x_{n+1} are designated as input and output signals, respectively. (b) Roadmap of 3D-MPST Block, dovetailing STrans with CNN modules. Note: Overlap Serialization and Volume Recombination are included to ensure the alignment of result dimensions from between STrans and CNNs.

undisclosed colon cancer, medical segmentation decathlon (MSD) brain tumor,²⁰ multiatlas labeling beyond cranial vault (BCV),²¹ and automated cardiac diagnosis challenge (ACDC),²² and (4) demonstrating the designed model not only outperforms those existing ones but also demands less parametrization and lower computational conditions in the metric of giga floating-point operations per second (GFLOPs), which illustrate the potency of our protocol for intricate MIS task handling.

THEORY AND COMPUTATIONAL METHOD

As depicted in Figure 1, the superstructure of the entitled parallelly multiscale Trans-CNN aggregation net-work is embodied as a U-form of encryptor-decryptor concatenation. Given inherent computing complexity in the self-attention mechanism of Trans, explicitly evaluating volumetric inputs by 3D-MPST would be infeasible. Thus, we started with methodical dimension decrease by use of pure convolutional block (Pure Conv Block) plus downsampling,^{23–25} for the sake of low-level but high-resolution extracts of fine-grained details. Likewise, the upsampling pathway mirrors these blocks at corresponding stages to restore image to its original dimensions.

In order to treat information in a high-fidelity fashion, both the encoder and decoder lines are skip-connected and carry out concatenation for CNNs, whereas superposition for Trans. This flexible networking endows the entire model with perplexing pattern-learning capability. On the other hand, a deep supervision mechanism is introduced to lift the learning efficiency, resulting in robust performance across multiple output levels. Afterward, the decoder effectively tackles and

transforms feeds from the encoder through upsampling and additional Pure Conv Blocks, and culminates quadruplexing outputs. The overall loss is calculated using a joint loss functional (including cross-entropy and dice loss).

Architectural Details of MPSTrans

First of all, to abate the anisotropy of 3D medical images, a downsampling strategy was enacted that reminiscent of nnU-Net,²⁶ utilizing “stride” to regulate the downsampling degree across different dimensions. Specifically, the input volume is denoted as $x \in \mathbb{R}^{H \times W \times D}$, where H , W , and D represent height, width and depth, respectively, and are set to 192, 192, and 48. We define five-level strides for downsampling along the depth direction as $M_j = [1, 2, 2, 2, 1]$ and for the aspects of height and width as $N_j = [2, 2, 2, 2, 2]$. Then, the features for each block can be quantified as $f_i \in \mathbb{R}^{H_i \times W_i \times D_i \times C_i}$, in which $H_i = H/N_i$, $W_i = W/N_i$, $D_i = D/N_i$, and $C_i = M_i C$; in addition, $N_i = \prod_{j=1}^i N_j$ and $M_i = \prod_{j=1}^i M_j$, C stands for base channels.

Pure Conv Block. In the initial stages of MPSTrans, Pure Conv Blocks are deployed.^{23–25} As shown in Figure 2a, it consists of a series of layer normalization (Layer Norm), Gaussian error linear unit (GeLU), 3D convolution layer (Conv3D), and residual connection (expressed as $\oplus$, which is extra crucial, as it facilitates direct data streaming inside block, eluding gradient evanescence or explosion). As such, the input at each layer directly correlates with the output, thereby boosting the learning capacities with intact info relay across layers. The whole working principle can be briefly described as

$$x_{n+1} = F(x_n) + x_n \quad (1)$$

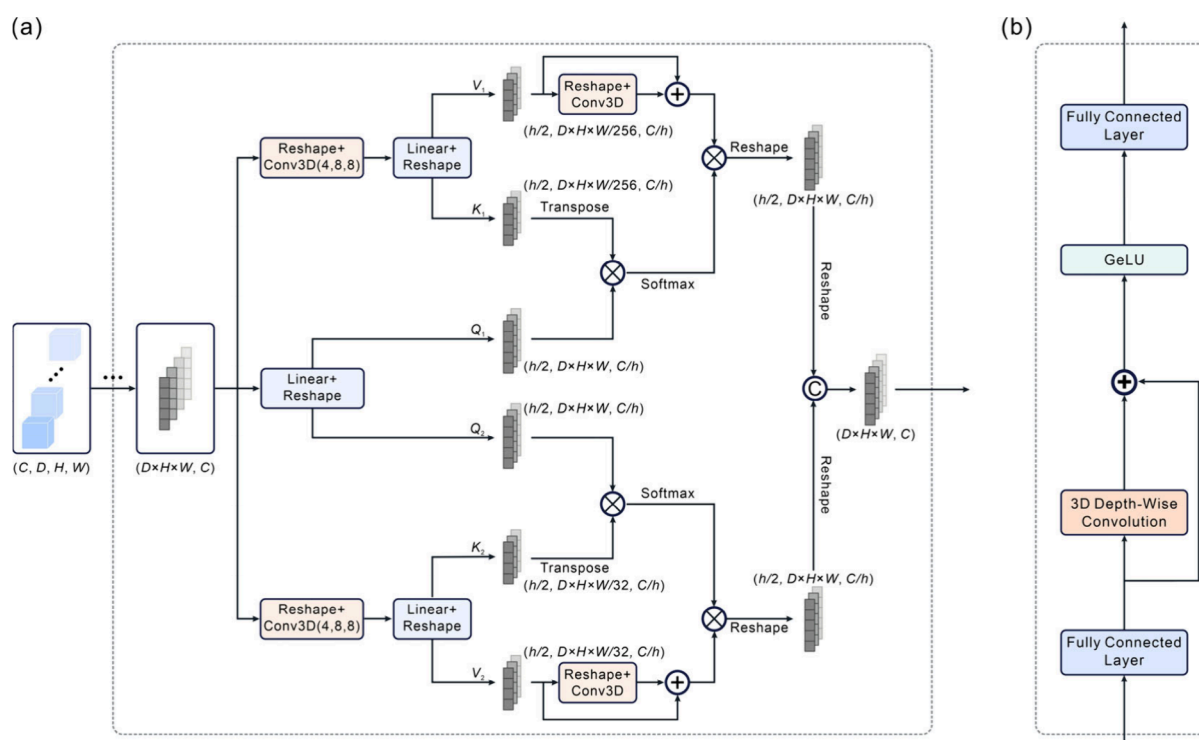


Figure 3. (a) Circuitry diagram of the 3D shunted self-attention mechanism. It begins with reshaping the input tensor; then the multiscale 3D self-attention is applied over multiheads, which downsamples keys and values to various extents, forming Q , K , and V tensors, and computes the attention using a Softmax operator to synthesize the results. (b) The modified feed-forward layer in the STrans module, which compared with the traditional one integrates a 3D depthwise convolutional layer between two fully connected layers.

where $F(\cdot)$ means the function of Pure Conv Block without $\oplus$ and x_n and x_{n+1} are implicated as the signal input and output, respectively.

3D-MPST Block. To pace up photoprocessing and attribute analysis, 3D-MPST Blocks, as dissected in Figure 2b as a pair of STans and CNN modules, are created to receive in a synchronous style the scaled signature maps from Pure Conv Blocks. The STans cell initiates with a key operator Overlap Serialization (O.S.) and ends with another, Volume Recombination (V.R., Figure 2b left), for volume and sequence transformation, which ensures a consistent output dimension with that CNN-generated. The CNN cell begins with Conv3D to expand dimensions, followed by DWConv3D (depthwise convolutional layer), a feature extractor, and concludes with another Conv3D for dimensionality decline (Figure 2b right). The consequences from CNN and STans are combined using $\oplus$. The entire encryption can be summarized as

$$y_i = V. R. \{ST^2[O. S. (x_{i-1})]\} + \text{Conv}^2(x_{i-1}) \quad (2)$$

where x_{i-1} is defined as the downsampled outcome from the $(i - 1)$ th encoding stage and ST^{22} and Conv^{22} indicate two executions of STans and CNN modules, respectively. Accordingly, the description of the decryption can be formulated as

$$z_i = V. R. \{ST^2[O. S. (x_{i+1} + y_i)]\} + \text{Conv}^2([x_{i+1}, y_i]) \quad (3)$$

where x_{i-1} and y_i are defined as the upsampled outcomes from the $(i + 1)$ th decoding and the i th encoding stages, respectively, and the notation $[x_{i-1}, y_i]$ delineates the concatenation of x_{i-1} and y_i along the channel dimension.

3D Shunted Self-Attention Layer. To address the special specification of 3D MIS, the standard Shunted Self-Attention was evolved into a 3D upgrade, as unveiled in Figure 3a. This conversion enables self-attentional computation within a single layer of STans, whereby the foregoing input $x \in \mathbb{R}^{H \times W \times D \times C}$ is reconfigured into $N \times D$ ($N = H \times W \times D$) to flatten 3D spatial dimensions while conserving channel information. The reshaped sequences are further projected into query (Q), key (K), and value (V) tensors to facilitate multihead (H of distinct attention heads) self-check mechanism in parallel, therefore enhancing the capture of diverse in-depth spatial relationships. To be specific, K and V are downsampled to various sizes suitable for the scale targeted by an i -indexed head as

$$Q_i = X \cdot W_i^Q \quad (4)$$

$$K_i = \text{MTA}(x, r_i) W_i^K \quad (5)$$

$$V_i = \text{MTA}(x, r_i) W_i^V + \text{DW}[\text{MTA}(x, r_i) W_i^V] \quad (6)$$

where $\text{MTA}(x, r_i)$ refers to the multiscale token aggregation layer in the i th head with a downsampling rate r_i , parameters (W_i^Q, W_i^K, W_i^V) weigh the linear projection in the i th head, and $\text{DW}(\cdot)$ is specialized for the depthwise convolution operation.

Next, to realize this downsampling, we adopt a Conv3D with a kernel size and stride of $(r_{\text{adp}}, r_i, r_i)$, where r_{adp} is dynamically adjusted based on the ratio of H or W to D of the input image. The variability of r_i across different heads allows K and V to attend to alterable spatial scope or scale of sequence input, therefore favoring catching comprehensive and complementary aspects. This tailored tactic operates optimal head-processing in the region of interest (ROI) on specific scales, which results

in promoted performance and pliability. Mathematically, the self-attention is given by

$$\text{Attention}(Q_i, K_i, V_i) = \text{Softmax}\left(\frac{Q_i K_i^T}{\sqrt{d}}\right) V_i \quad (7)$$

where d is deemed as the dimension. As illustrated in Figure 3b, we complement the classic feed-forward layer by inserting DWConv3D between two fully connected layers to better bring in local details, which can be described as

$$x'_{\text{in}} = \text{FC}(x_{\text{in}}) \quad (8)$$

$$x_{\text{out}} = \text{FC}[x'_{\text{in}} + \text{DW}(x'_{\text{in}})] \quad (9)$$

where $\text{FC}(\bullet)$ corresponds to the connected layer, x_{in} denotes the down- or upsampled signature and x_{out} the resulting output.

Independent Down- and Upsampling Layers. Distinct down-/upsampling layers are incorporated between the en-/decoding stages. To tone training stability, Layer Norm is recruited:²⁷ before the downsampling layer to halve feature map size so as to double the channels and after the upsampling layer to double the size of the feature map and accordingly cut down channels by half. Provided the nonuniform nature of 3D medical images, the resampling stride is by default settled to 1 in anisotropic directions, which maximizes essential feature fetches, in the meantime sustaining spatial info integrity throughout the segmentation.

Deep Supervision Mechanism. To maximize multiscale feature finding, deep supervision steps into the upsampling stages of the decoder. Supervisory signals are introduced in these threads to speed convergence while slowing gradient disappearance, which facilitates feature extractions across scalable space resolution. A weighted Loss Function is functioned to trade the dedications off from different resolutions. To be specific, lower-resolution layers, which withhold more macroscopic information, are imposed with less weight, typically in the range of 0.1–0.3, while higher-resolution ones, which do more microscopic information, are with larger weights ranging from 0.5 to 1.0. This scheme secures that the model prioritizes perusing local details, while still skimming the global landscape, ultimately uplifting segmentation accuracy by allowing for both proximal and distal distinctions.

Loss Function. The cross-entropy loss function (L_{CE}) is in conjunction with the dice loss (L_{Dice}).^{17,26} The former, commonly for classification tasks, renders a robust measure of appraisal; the latter handles class imbalance and holds spatial continuity. Together, they are intended to tune segmentation and classification accuracies as well as spatial coherence, with equations elaborated as

$$L_{\text{CE}} = -\frac{1}{k} \sum_{i=1}^k [y_i \log y'_i + (1 - y_i) \log(1 - y'_i)] \quad (10)$$

$$L_{\text{Dice}} = \frac{1}{k} \sum_{i=1}^k \left(1 - 2 \times \frac{y'_i \cap y_i}{y'_i + y_i}\right) \quad (11)$$

$$L_{\text{cascade}} = \frac{1}{2} L_{\text{CE}} + \frac{1}{2} L_{\text{Dice}} \quad (12)$$

where y_i and y'_i delimit the ground truth mask and the prediction one, respectively, and L_{cascade} specifies the total loss function that trained to optimize the model.

Multimodal Image Segmentation

This was undertaken for [1] an undivulged colon cancer plus three public data sets involving [2] BCV,²⁸ [3] MSD brain tumor,²¹ and [4] ACDC;²² both Dice similarity coefficient (DSC) and Hausdorff distance at 95% (HD95) were applied as the key evaluators.

For [1], 79 annotated PET-CT bimodal images of colorectal cancer were obtained from Nanjing Drum Tower Hospital. Of these, 70 samples were assigned for training and model optimizing, while 9 were reserved for testing. To standardize the data, the PET specimens were scaled to the CT resolution (512×512), and each pristine image order was reformatted to the Neuroimaging Informatics Technology Initiative, for the best of volumetric partition preparation. For [2], it contains 30 cases, up to a total of 3779 axial abdominal clinical CT pictures targeting eight viscera (aorta, gallbladder, pancreas, spleen, stomach, liver, left and right kidneys), which was divided into 18 for training and 12 for testing in compliance with the established methodologies.²⁹ For [3], it visualizes 484 brain tumors via MRI from 19 institutes and 2016–2017 “Brain Tumor Segmentation Challenge”,³⁰ each presented in profiles of fluid-attenuated inversion recovery, T1-weighted, T1-weighted postgadolinium, and T2-weighted, with primary ROIs at the subregion of edema, (non)enhancing tumor, and core. The fractions of training, validation, and test subsets are 80%, 15%, and 5%, respectively.¹³ For [4], it constitutes cardiac MRI data from 100 patients, with segmenting targets on left and right ventricles and myocardium. The ratio of subdivisions is 70 (training):10 (validation):20 (test).

Implementation Details

MPSTrans was run on Ubuntu 22.04 using Python 3.9.16 and PyTorch 2.0.0. Training was conducted on a single NVIDIA 3090 GPU equipped with 24 GB of memory. For a fair comparison, all 3D segmentation experiments were also conducted in nnU-Net framework. The initial training was hyperparametrized in accordance with those previous preliminary studies,²⁶ setting the learning rate to 0.01 for CNN layers and 0.001 for STrans modules to guarantee gradual grasp of global traits, with a momentum of 0.99 and a weight decay of 3×10^{-5} using the Stochastic Gradient Descent optimizer. For 3D-MPST Blocks, the self-attention heads, the downsampling rate r_i , and the base number of channels were settled to [2, 4, 8, 16], [8, 4, 2, 1], and 16, respectively. In pursuit of training robustness, we resorted to eq 12; the training period spanned over 1000 epochs, each composed of 250 iterations. The stride strategy of nnU-Net was used for both down- and upsamplings. During training, we randomly cropped subvolumes from the raw data, to serve as inputs to our model as compiled in Table 1.

Table 1. Subvolumes and Batch Sizes Cropped from Corresponding Datasets during Training

Data Set	Sub-Volume	Batch Size
BCV	$48 \times 192 \times 192$	2
ACDC	$16 \times 224 \times 224$	2
MSD Brain Tumor	$128 \times 128 \times 128$	2

Table 2. Hyperparametric Seeking Scopes and Selections

Hyperparameter	Initial Setting	Random Search Range	Final Value	Grid Search Range	Final Value
Learning Rate (CNN)	1×10^{-2}	1×10^{-4} to 1×10^{-2}	5×10^{-3}	5×10^{-4} to 5×10^{-3}	1×10^{-3}
Learning Rate (STrans)	1×10^{-3}	1×10^{-5} to 1×10^{-3}	1×10^{-4}	1×10^{-5} to 1×10^{-4}	1×10^{-4}
Weight Decay	3×10^{-5}	1×10^{-5} to 1×10^{-3}	1×10^{-4}	1×10^{-4} to 3×10^{-4}	3×10^{-4}
Momentum	0.99	[0.90, 0.95, 0.99]	0.99	[0.95, 0.99]	0.99
Dropout Rate	0.1	[0.1, 0.2, 0.3, 0.4]	0.2	[0.2, 0.3, 0.4]	0.2

Further, hyperparameters were honed with the BCV data set. To screen out the optimal configuration, random and gridding search strategies were wielded to weigh key factors (such as learning rate, weight decay, momentum, and dropout rate), with performance appraised on a separate set of validation to retain rigorous generalization and grant no overfitting. As a start, a coarse-grained stochastic search was launched over 20 randomly sampled setups of hyperparameters with a uniform distribution in predefined ranges. Proceeding from these findings, we refined and narrowed the hyperparametric ranges, then optimized them in a fine-grained grid. The final hyperparameters were determined with the best trade-off between DSC and HD95. Table 2 lays out the optima in multiranges.

The quantitative consequences in Table 3 testify that the systematic 2-stage hyperparameter honing strengthens the

Table 3. Segmentation Statistics under Different Search Strategies

Search Strategy	Average	
	HD95↓ (mm)	DSC↑
Initial Setting	9.634	0.865
Random Search	8.915	0.873
Grid Search	8.405	0.881

segmentation performance profoundly. Notably, the final hyperconditions obtained through Grid Search struck a 12.7% deduction in HD95 and a 1.85% improvement in DSC compared to the Initial Setting.

RESULTS AND DISCUSSION

Comparison among State-of-the-Art Approaches

In this section, the performance of MPSTrans is compared with a variety of prior methods, which include both 2D segmentation techniques of U-Net,³¹ SwinU-net,³² TransU-Net,^{15,16} AttU-Net,^{33,34} and MissFormer³⁵ and 3D segmentors of 3D U-Net,³⁶ nnU-Net,²⁶ PHTrans,²⁷ UNETR, and Swin UNETR.¹³ In terms of fairness, all 3D models were retrained under consistent experimental conditions.

A newly constructed (unpublished) colon cancer data set was exemplified to manifest the performance of MPSTrans against other 3D models, with results recorded in Table 4, wherein 3D U-Net was built on the general U-Net by replacing 2D convolutions with 3D versions. This modification limited its specialty to grab global information by crescendo computational resources. Moreover, both nnU-Net and PHTrans rely on multilayered convolutions for feature extraction, which compromise efficiency, as suggested by their high FLOPs. Additionally, their heavy reliance on convolutions may prevent precise segmentation of organ boundaries. In contrast, MPS-Trans availed paralleled processing with the help of 3D-MPST Block, as the fusion of STrans

Table 4. Segmentation Scores on Colon Cancer Dataset Using the Metrics of HD95 and DSC^a

Method	Params. (M)	FLOPs (G)	Average	
			HD95↓ (mm)	DSC↑
3D U-Net	58.247	652.459	11.815	0.641
nnU-Net	31.224	289.938	5.544	0.838
PHTrans	37.196	173.826	5.213	0.839
Swin UNETR	69.936	378.977	6.741	0.763
MPSTrans	41.744	75.342	3.826	0.859

^aBoldface values hint at the best.

with CNN greatly aggrandizes the information flow efficacy and feature-extracting facility (Table 4), which highlights superb segmentation power with an average HD95 of 3.826 mm, an average DSC score of 0.859, and an economical FLOP down to 75.342 (in bold).

The visualization results from the data archive of colon cancer clearly corroborated that MPSTrans was superior to other contenders (Figure 4). Although PHTrans behaved better than Swin UNETR, nnU-Net, and 3D U-Net, it was prone to obvious oversegmentation issues. On the contrary, MPSTrans achieved remarkable accuracy, displaying a high degree of overlap with truly tagged PET-CT and CT shots. Other algorithms revealed inconsistencies more or less in their outcomes. For instance, although nnU-Net, renowned for its adaptive architecture, outflanked 3D U-Net in several respects, it was still faced with challenges in gleaning fine-grained details, especially at axial locations. In other words, its flexibility in determinants (i.e., layer depth and kernel sizes) could not compensate for the shortage of a stationary solution to garnering global/local features from the entanglements of e.g. a colon-cancer data set. Aside from that, 3D U-Net failed to offer effective reconstitutions from any perspective, since it was trained following a fixed framework of manual hyperparametrization, that is inferior to nnU-Net in self-adaptation, indicative of the limitation in seizing L&G information. Apart from that, though Swin UNETR managed to rebuild some volume of the volumetric graphics, it overused self-attentions and struggled to apprehend axial nuances, hence being less eligible for global context consolidation.

In short, these observations emphasize the merits of MPSTrans in detail discriminability, which is accredited to its strategic handshake with self-attentive mechanisms in an optimal network. Collectively, these elements empower MPSTrans to excel in MIS missions.

Performances of State-of-the-Art Approaches

To verify the versatility of MPSTrans for processing other types of medical images, three open access databanks, BCV, MSD and ACDC, were inspected with a summary of segmentation performance against other models in Tables 5–7. The findings confirm that 3D modeling outshines that of 2D as pertains to HD95 and DSC, which is mainly ascribed to

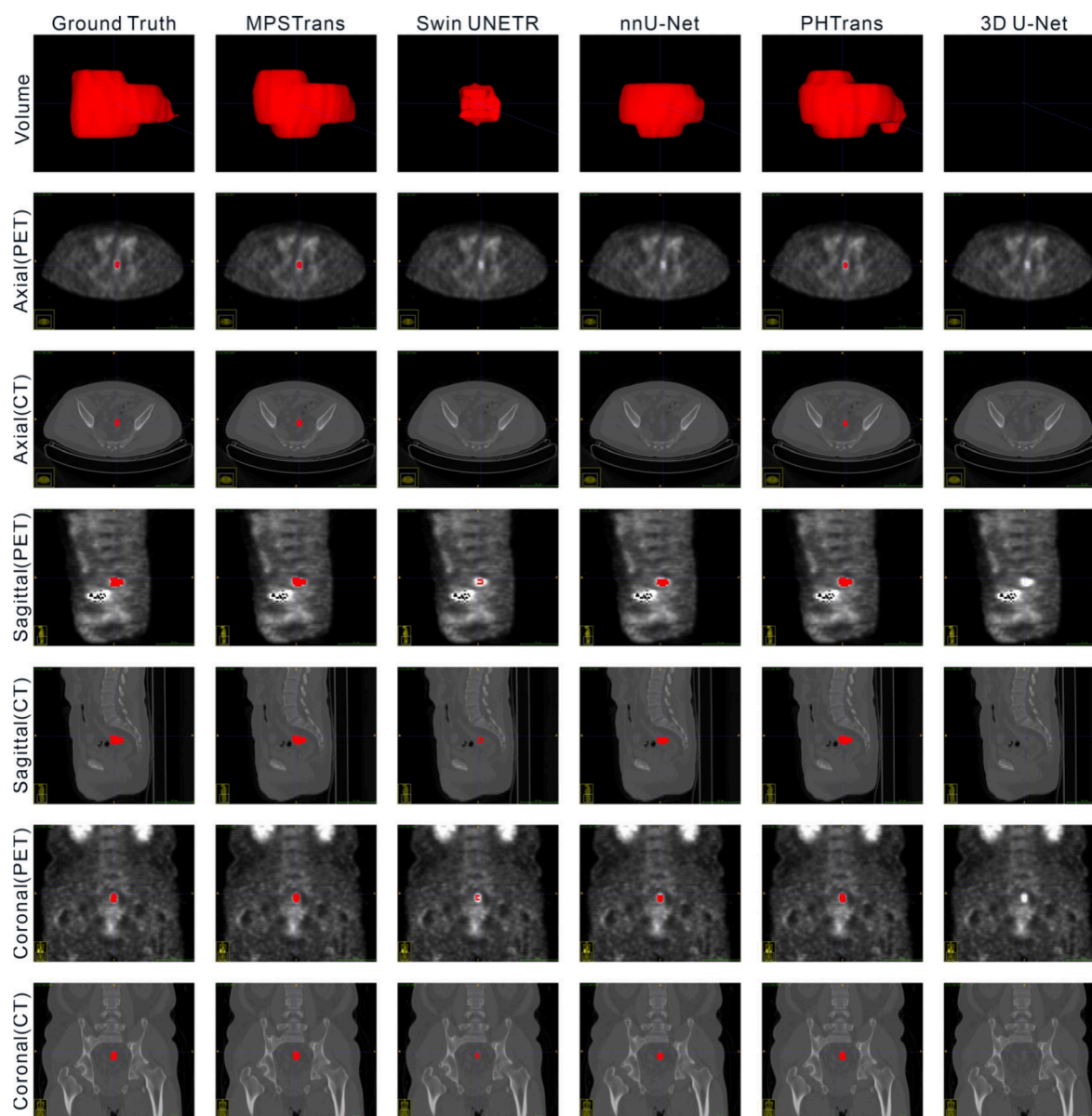


Figure 4. Qualitative and comparative results from segmenting colon cancer database. The panel aligns an output array in (from top to bottom) volume, axial, sagittal, and coronal PET/CT views for (from left to right) Ground Truth, MPSTrans, Swin UNETR, nnU-Net, PHTrans, and 3D U-Net.

the ability of 3D informatic retention, leading to more comprehensive and accurate segmentations. Among all 3D candidates, nnU-Net and PHTrans exhibit close exactitudes on BCV and MSD. Nevertheless, PH-Trans attains approximately 66.5% and 34.6% greater computational efficiency as shown in Table 5 (188.203) and Table 6 (176.363), respectively, which should be attributed to its convolutional and self-attentive kernels. The proposed MPSTrans embeds a 3D-MPST Block for Conv. operational optimization and computational redundancy reduction, resulting in efficiency gains of about 56.4% and 55.8% (Table 5 and Table 6, bold). More than that, in BCV (Table 5 and Table 6), MPSTrans prevails over PHTrans in the suboptimal in both HD95 ($8.405 < 11.128$ mm) and DSC (0.881 vs. 0.880). Similarly, in ACDC (Table 7), MPSTrans has the highest DSC score as compared to that of nnU-Net; conversely, UNETR is fairly weak in segmentation, probably owing to its focus on global context while being short of localized finesse.

Further, we substantiated the segmentation solution by MPSTrans with those by Swin UNETR, UNETR, nnU-Net, PHTrans, and 3D U-Net as arranged in Figures 5–7. MPSTrans matches well with the Ground Truth labels in innards demarcation, particularly in predicting perimeter and basal plane of tissues with a DSC of 0.859. Such coincidence rightly reflects the knack of 3D-MPST in cross-scale characteristic acquisition (Volume row in Figure 5). Whereas from Swin UNETR to 3D U-Net, these tools seem fallible in outlining the aorta and contouring a complete stomach. In regard to other views (Axial, Sagittal, Coronal in Figure 5), MPSTrans also stands out in distinguished differentiability. On the other side, UNETR etc. possess poor preciseness in segmentation ($DSC \leq 0.792$), resulting in organic confusion and regional misjudgment of gallbladder, liver, spleen, and stomach.

Getting down to the exploration of brain tumor and cardiac structure segmentations, the MPSTrans model keeps up the good work, as shown in Figure 6, and performs exceptionally

Table 5. Segmentation Scores on BCV Dataset Using the Metrics of HD95 and DSC^a

				Average	
				HD95↓ (mm)	DSC↑
2D	Method	Params. (M)	FLOPs (G)		
	U-Net			10.212	0.664
	AttU-Net			9.956	0.665
	TransU-Net			13.006	0.644
3D	3D U-Net	58.201	704.510	14.580	0.859
	nnU-Net	30.812	313.613	11.115	0.875
	PHTrans	36.404	188.203	11.128	0.880
	UNETR	92.608	82.702	31.600	0.792
	Swin UNETR	69.604	408.533	14.930	0.858
	MPSTrans	32.815	81.905	8.405	0.881

^aBoldface values hint at the best.**Table 6. Segmentation Scores on MSD Brain Tumor Dataset Using the Metrics of HD95 and DSC^a**

				Average	
Method		Params. (M)	FLOPs (G)	HD95↓ (mm)	DSC↑
2D	U-Net			10.212	0.664
	AttU-Net			9.956	0.665
	TransU-Net			13.006	0.644
3D	3D U-Net	58.223	837.006	4.305	0.761
	nnU-Net	31.210	269.866	4.336	0.768
	PHTrans	37.257	176.363	4.800	0.767
	UNETR	102.114	101.729	5.126	0.754
	Swin UNETR	69.906	487.718	5.536	0.754
	MPSTrans	38.443	77.971	4.577	0.778

^aBoldface values hint at the best.**Table 7. Segmentation Scores on ACDC Dataset Using the Metric of DSC^a**

	Method	Params. (M)	FLOPs (G)	Average DSC↑
2D	SwinU-Net			0.900
	TransU-Net			0.897
	MISSFormer			0.908
3D	3D U-Net	58.222	319.426	0.890
	nnU-Net	30.215	155.006	0.912
	UNETR	92.669	37.535	0.821
	Swin UNETR	69.933	185.116	0.886
	MPSTrans	31.115	47.527	0.915

^aBoldface values hint at the best.

well for precisely predicting edema-associated nonenhancing ROIs. Regardless, there is still some deviation between the predicted and the authentic. Such a sign of overprediction pinpoints that enhanced edge examination is necessary. In comparisons from varying view angles, MPSTrans wins at its insightful detail discovery, which plays pronouncedly in the Axial view, where its pinpoint prediction of cerebral edema in the lower right gives it an eminent edge over Swin UNETR, nnU-Net, and other incompetents (by the way, UNETR and

3D U-Net expose overpredicting instances). Albeit PHTrans acquires analogous appearances in this domain, it hardly succeeded in the identification of edematous status.

As for cardiac structural segmentation (Figure 7), MPSTrans also bests the rivals in delicate detail discernibility in volumetric critical zones such as right ventricle and left ventricular myocardium/cavity. Others encounter issues with omission or oversegmentation in certain classifications. In the Axial line, MPSTrans coincides closely with the true tracing on the left, while other partitioners display drastic discrepancies in specific spaces. For example, only MPSTrans can unerringly recognize the right region of a ventricle from background, an adept that was difficult for other models to match. Moving onto the Sagittal and Coronal, MPSTrans upholds high uniformity with the Ground Truth group, as compared with mistakes made by others in ROIs. In a nutshell, whether in brain or heart tissue treatment, MPSTrans outperforms those mainstream models, involving Swin UNETR, UNETR, nnU-Net, PHTrans, and 3D U-Net, with its special skills at satisfactorily segmenting both sidelines and shapes of ROIs.

By referencing to other models, it was shown that MPSTrans accomplished significant segmentation specificity toward readsets, regardless of its fewer FLOPs and variables. This apparent appraisal is not only because of the change in capacity, but rather the novelty in tectonics by bridging CNN with STans: CNN concentrates on withdrawing details around regional rims and fabrics, whereas Trans distills interdependencies in global vision via self-attentions. Such a marriage into MPSTrans endeavors to equilibrate their bifunctions but bests both (in brief, “1 + 1 > 2”). Moreover, MPSTrans escalates efficiency by parallelly processing CNN and Trans inside its 3D-MPST block, enabling effectual fingerprint fetching. Unlike traditional 3D U-Net models that support single-type computations at each layer, MPSTrans evades excessive calculations by adapting the parallelized paradigm for reducing repetitive retrieval of features, which in fact is privileged by its parental Trans. This construct underlies the MPSTrans competency in curtailing computational and parametric perplexity, thus striking satisfactory MIS.

Ablation Study

In balancing L&G feature extraction, MPSTrans combines CNN and STans modules in parallel. The CNN component engages extracting edges and textures, while the Trans partner pays attention to catch contextual correlations, e.g., visceral multidimensional dependencies. Such complementarity makes MPSTrans compartmentalize medical images in a more meticulous manner. To elucidate contributions of CNN and Trans branches in the 3D-MPST Block, an ablation study was used for abdominal multisplanchnic segmentation on the BCV repository as given in Table 8. In these trials, either the CNN or Shunted Trans part was suspended, by which their impact was deliberated on the modified model as Model #1 (i.e., MPSTrans w/o CNN) and #2 (MPSTrans w/o STans). It is worth noting that the two were tweaked for parametric counts on par with those of the untampered MPSTrans. Such an investigation yielded six major insights.

(1) Model #2 outweighs #1 in partition performance, particularly in respect to the HD criterion (13.604), which advantage attests to the authenticity of CNN in local fingerprinting. (2) Model #1 outshines in fewer FLOPs than #2 (91.346), which is contributed by the global dependency detectability of the self-attentive STans, that pays attention to

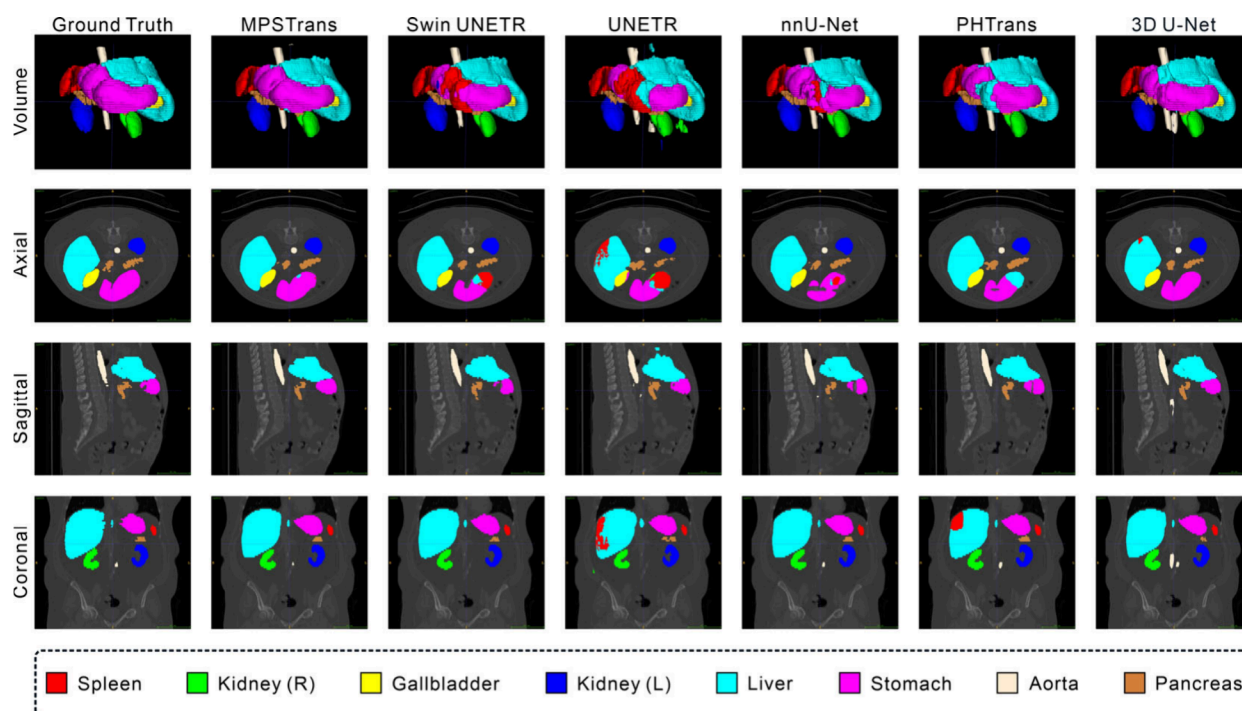


Figure 5. Qualitative comparison of BCV-based abdominal multisplanchnic segmentation sequences in (from top to bottom row) Volume, Axial, Sagittal, and Coronal views among (from left to right column) Ground Truth, MPSTrans, Swin UNETR, UNETR, nnU-Net, PHTrans, and 3D U-Net models.

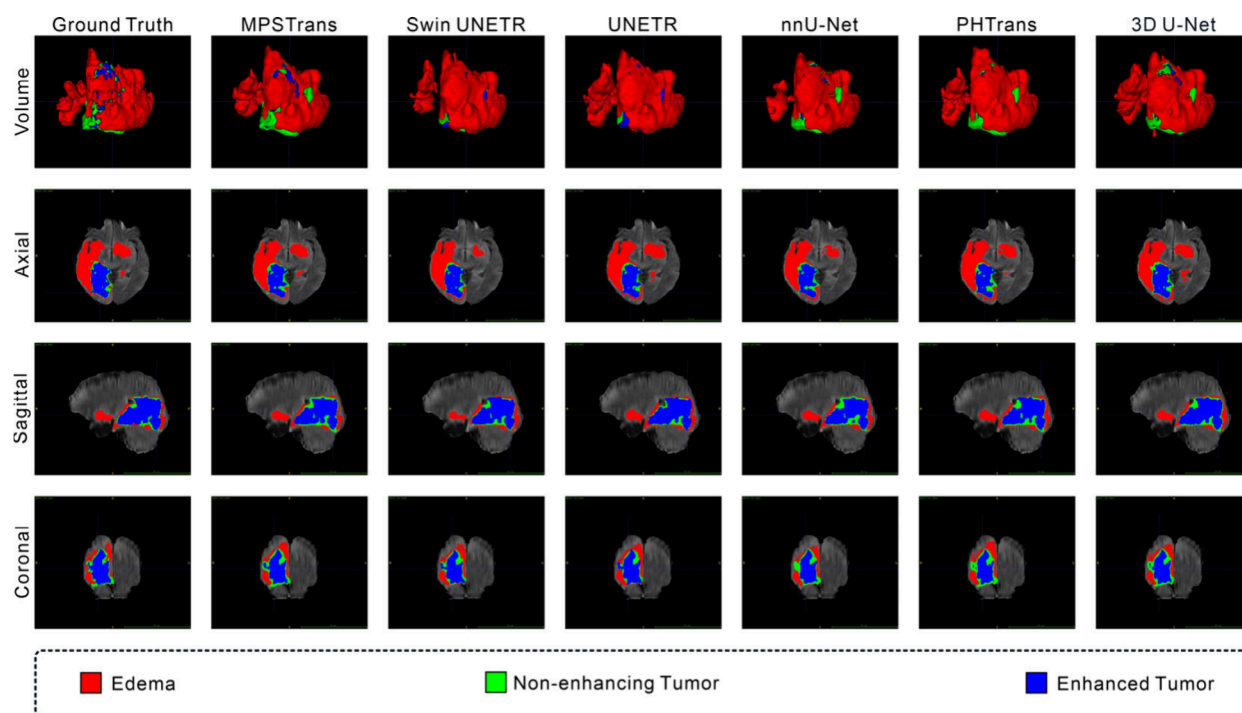


Figure 6. Qualitative comparison of MSD-based brain tumor segmentation sequences in (from top to bottom row) Volume, Axial, Sagittal, and Coronal views among (from left to right column) Ground Truth, MPSTrans, Swin UNETR, UNETR, nnU-Net, PHTrans, and 3D U-Net models.

the pivotal parts of single-layer inputs. (3) Model #1 shows a surge in HD95 (15.008) and a smaller DSC (0.870) as contrasted to MPSTrans. The removal of the CNN constituent caused losses of localized subtlety, which delivered difficulty in deliberating details, especially at the interface of entrails. Hence, it begot a bad underperformance. (4) Model #2 also appears outcompeted, with HD95 ascending to 13.604 and

DSC descending to 0.875. In spite of this, it indeed perceived imperceptible changes; it omitted the overview, which hampered handling long-range or large-scale labyrinths. (5) Appreciably, the two derivatives (as *apo*-MPSTrans) do run near *holo*-MPSTrans, but they still lag behind in strengths of segmentation and calculation (bold), which gap arises because MPSTrans not only optimized local character captures but also

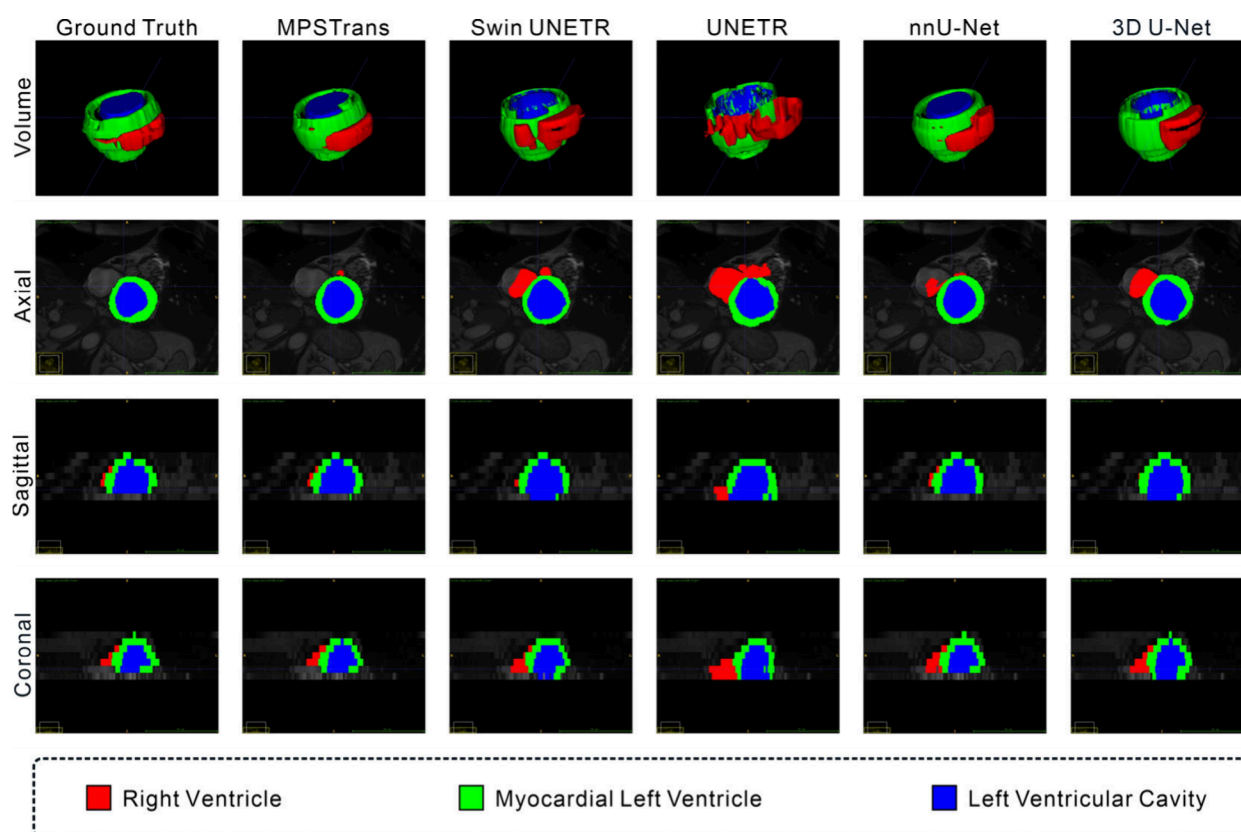


Figure 7. Qualitative comparison of ACDC-based heart structure segmentation sequences in (from top to bottom row) Volume, Axial, Sagittal, and Coronal views among (from left to right column) Ground Truth, MPSTrans, Swin UNETR, UNETR, nnU-Net, and 3D U-Net models.

Table 8. Ablation Assay on MPSTrans and Its Branch-Removed Controls Using the Metrics of HD95 and DSC

Method	Params. (M)	FLOPs (G)	Average	
			HD↓ (mm)	DSC↑
MPSTrans w/o CNN branch Model #1	31.729	91.346	15.008	0.870
MPSTrans w/o STans branch Model #2	28.411	154.483	13.604	0.875
MPSTrans	32.808	81.918	8.415	0.881

manipulated multiscale messaging in a paralleled pipeline. (6) MPSTrans, being benefiting from the binary of both CNN and Trans, overshadows other settings. This predominance implies the important copresence of L&G extractors for a full-fledged segmentator, which successfully overcame the shortcomings of Models #1 and #2 (not just magnifying the maximum distances but diminishing the overlaps between artificial and actual labels). Consequently, this coordination saved computational cost, while speeding up the info circulation.

Last but not least, irrespective of the comparable Params counts among Model #1 (31.729 M), Model #2 (28.411 M), and MPSTrans (32.808 M), the first two models require around 12% and 88% more FLOPs than the last. This underscores that the MPSTrans outperformance does not stem from the amount of parameters but rather originates from its innovative infrastructure in conjugating CNN with Trans, which not only enhanced the exactness of segmentation but also ameliorated computational complication, altogether affirming again the asset of MPSTrans.

CONCLUSION

In this work, a new network MPSTrans is contrived to thoroughly task with 3D-MIS. By orchestrating the regime of STans and CNN, it exploits multiscale slicing to assist in feature familiarization and analyze visual variation for performance progress. Expansive estimations across *meta* medical data sets demonstrated that MPSTrans overwhelmed existing models in both segmenting sophistication and computational load, especially in the scrutiny of small tumor areas with dramatically dwindled FLOPs by 56.4% for BCV, 71.1% for MSD, 69.3% for ACDC, and 56.7% for colon cancer, which proved promising for clinical deployment. Being aware of that, its reliability relies on the availability of adequate annotated data, which might be inaccessible in resource-restrained scenarios; future efforts will be invested in semisupervised learning or synthetic data production to aggrandize applicability. After all, MPSTrans holds plenty of prospects to advance the adaptation of Trans-based segmentators for biomedical imageology.

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Notes

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